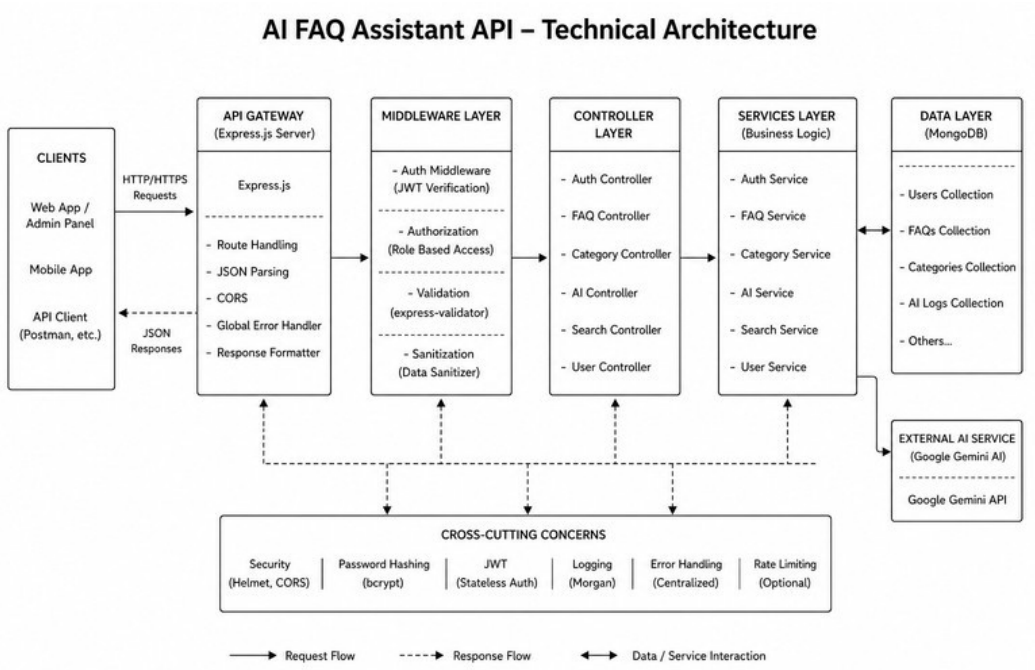
Solution Architecture

Date	17 September 2026
Team ID	SWTID-2026-3976
Project Name	AI FAQ Assistant API (AnswerBot)
Maximum Marks	5 Marks

Solution Architecture Diagram:

The solution follows a layered architecture: clients call the Express API Gateway, requests pass through middleware (JWT verification, validation, error handling), controllers hand work to the service layer, and services use MongoDB and the external Google Gemini API.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation / Client Layer	API clients that send HTTP/HTTPS requests and receive JSON (Postman, web or mobile apps).	Postman, ThunderClient, any HTTP client
API Gateway (Express Server)	Handles routing, JSON parsing, CORS and the global error handler.	Node.js, Express.js, cors, dotenv
Middleware Layer	Verifies JWT tokens, validates and sanitises input, formats errors.	jsonwebtoken, validationMiddleware, errorMiddleware
Controllers and Services	Controllers for auth, FAQ and AI; service layer holds business logic and the Gemini integration.	Express controllers, geminiService.js
Database	Stores users and FAQs with schema validation.	MongoDB Atlas, Mongoose

Component Name	Description / Role in Architecture	Technologies Used
External AI Service	Generates answers and complete FAQs from a topic or question.	Google Gemini API (@google/genai)

Data Model (implemented collections)

Collection	Fields
User	_id, name (required), email (required, unique), password (hashed, min 6 characters), role (default "user"), createdAt, updatedAt
FAQ	_id, question, answer, category (Technology / Education / Health / Banking / General), createdBy (reference to User), createdAt, updatedAt

Relationship: one User can create many FAQs (User 1 : N FAQ). The full ER diagram, which also shows planned entities (Category, Tag, FAQ_LIKE, ANSWER_FEEDBACK), is included in the Project Documentation.